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FLEX CIRCUIT AROUND CORE WIRE IN INTRALUMINAL DEVICE AND ASSOCIATED DEVICES, SYSTEMS, AND METHODS

Abstract

An intraluminal device includes a flexible elongate member configured to be positioned within a body lumen of a patient. The flexible elongate member includes a distal portion comprising a distal core member and a sensor. The distal core member includes a length between a distal end and a proximal end. The flexible elongate member includes a proximal portion comprising a proximal electrical conductor. The flexible elongate member includes a flex circuit positioned around a majority of the length of the distal core member. The flex circuit includes a distal portion coupled to the sensor and a proximal portion coupled to a proximal electrical conductor such that the proximal electrical conductor is in electrical communication with the sensor via the flex circuit.

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Background/Summary

TECHNICAL FIELD

[0001] The subject matter described herein relates to intraluminal physiology sensing devices (e.g., an intravascular pressure sensing and/or flow sensing guidewire). For example, the intraluminal device may include a flex circuit for providing power and/or data communication between a distal sensor and a proximal electrical connector.

BACKGROUND

[0002] Existing intravascular guidewires with a sensor have fine-gauge electrical wires that provide transmission of electrical signals for the sensor. Manufacture of such devices may include many steps that involve human operator or machine contact with the fine-gauge electrical wires. Because of their delicate nature, the fine-gauge electrical wires are prone to damage as a result of such contact. For example, the conductors themselves may break and the insulation may be damaged. This leads to poor or no electrical connectivity for the sensor. Additionally, the fine-gauge electrical wires typically have circular cross-section, which occupies space with the very small outer diameter of the guidewire.

[0003] The information included in this Background section of the specification, including any references cited herein and any description or discussion thereof, is included for technical reference purposes only and is not to be regarded as subject matter by which the scope of the disclosure is to be bound.

SUMMARY

[0004] Disclosed are intraluminal physiology sensing devices (e.g., an intravascular pressure-sensing and/or flow-sensing guidewire) that include flex circuits for power and signal communication. The flex circuit may be formed separately and then applied to one or components of an intraluminal physiology sensing device, which permits improved electrical/mechanical performance and reduce or eliminate damage associated with handling the fine-gauge electrical wires. Because flex circuits have a flatter profile, flex circuits allow for a larger outside diameter (OD) distal core wire and/or proximal core wire and eliminate damage/scrap due to multi-filar bundle handling, which may enable lower electrical resistance/impedance along the length of the electro-mechanical intraluminal sensing device. Furthermore, the straightness and torque response of the intraluminal device with flex circuits may be improved as well because ribbons and/or multi-filar conductor bundles add stiffness and local torques to a guidewire or catheter.

[0005] The flex circuit assembly disclosed herein has particular, but not exclusive, utility for intraluminal medical catheters, guidewires, or guide catheters.

[0006] In an exemplary aspect, an intraluminal device is provided. The intraluminal device includes a flexible elongate member configured to be positioned within a body lumen of a patient, wherein the flexible elongate member comprises: a distal portion comprising a distal core member and a sensor, wherein the distal core member comprises a length between a distal end and a proximal end; a proximal portion comprising a proximal electrical conductor; and a flex circuit positioned around a majority of the length of the distal core member, wherein the flex circuit comprises a distal portion coupled to the sensor and a proximal portion coupled to a proximal electrical conductor such that the proximal electrical conductor is in electrical communication with the sensor via the flex circuit.

[0007] In some aspects, the flex circuit comprises: a polymer base; and a conductive trace formed on the polymer base. In some aspects, the flex circuit further comprises a polymer topcoat formed over the conductive trace. In some aspects, the flex circuit further comprises: a distal conductive pad formed on the polymer base at a distal portion of the flex circuit; and a proximal conductive pad formed on the polymer base at a proximal portion of flex circuit, wherein the conductive trace is coupled to the distal conductive pad and the proximal conductive pad, wherein the proximal electrical conductor is coupled to the flex circuit at the proximal conductive pad, and wherein the sensor is coupled to the flex circuit at the distal conductive pad. In some aspects, an adhesive is positioned between the flex circuit and the distal core member, wherein the adhesive is configured to couple the flex circuit and the distal core member. In some aspects, flex circuit is wrapped in a spiral pattern around the distal core member. In some aspects, the flex circuit is rolled in a cylindrical configuration around the distal core member. In some aspects, the flex circuit extends parallel to a longitudinal axis of the flexible elongate member around the distal core member. In some aspects, the flex circuit is formed prior to being positioned around the distal core member. In some aspects, the flex circuit is formed as a sheet, wherein at least one conductive trace of the flex circuit is formed across the sheet, and wherein the sheet is wrapped around the distal core member. In some aspects, the proximal portion further comprises a proximal core member, wherein the proximal core member comprises a distal end and a proximal end, wherein the flex circuit is positioned around the proximal core member. In some aspects, the distal portion further comprises: a polymer coating positioned around the flex circuit. In some aspects, the polymer coating is in direct contact with the flex circuit. In some aspects, the distal portion further comprises: a hydrophilic coating positioned around the flex circuit. In some aspects, the hydrophilic coating is in direct contact with the flex circuit.

[0008] In an exemplary aspect, an apparatus is provided. The apparatus includes an intravascular guidewire configured to be positioned within a blood vessel of a patient, the intravascular guidewire comprising: a distal portion comprising a distal core member and a sensor, wherein the distal core member comprises a length between a distal end and a proximal end; a proximal portion comprising a proximal electrical conductor; and a flex circuit positioned around a majority of the length of the distal core member, wherein the flex circuit comprises a distal portion coupled to the sensor and a proximal portion coupled to a proximal electrical conductor such that the proximal electrical conductor is in electrical communication with the sensor via the flex circuit.

[0009] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit the scope of the claimed subject matter. A more extensive presentation of features, details, utilities, and advantages of the metal ink conductor assembly, as defined in the claims, is provided in the following written description of various embodiments of the disclosure and illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Illustrative embodiments of the present disclosure will be described with reference to the accompanying drawings, of which:

[0011] FIG. 1 is a diagrammatic top view of an intravascular device, according to aspects of the present disclosure.

[0012] FIG. 2 is a diagrammatic side view of an intravascular sensing system that includes an intravascular device, according to aspects of the present disclosure.

[0013] FIG. 3A is a schematic top view of an unrolled flex circuit for application to a portion of an

intravascular device with electrical conductors/traces formed in the flex circuit, according to aspects of the present disclosure.

[0014] FIG. 3B is a schematic end view of a rolled flex circuit for application to a portion of the intravascular device from direction P shown in FIG. 3A rolled in a cylindrical configuration along a longitudinal axis and a cross-sectional end view along section line A-A in FIG. 5, according to aspects of the present disclosure.

[0015] FIGS. 4A-4D are schematic side views of a portion of a flex circuit including electrical conductors/traces disposed onto a substrate of the flex circuit, according to aspects of the present disclosure.

[0016] FIG. 5 is a diagrammatic side view of a portion of an intraluminal device **102** with electrical conductors/traces **328** formed in the flex circuit **320**, according to aspects of the present disclosure.

[0017] FIG. 6 is a diagrammatic cross-sectional side view of a portion of an intraluminal device **102** with electrical conductors/traces **328** formed in the flex circuit **320**, according to aspects of the present disclosure.

[0018] FIG. 7 is a diagrammatic cross-sectional side view of a portion of an intraluminal device **102** with electrical conductors/traces **328** formed in the flex circuit **320**, according to aspects of the present disclosure.

DETAILED DESCRIPTION

[0019] Disclosed is a flex circuit assembly that provides an improved electrical/mechanical performance for an intraluminal sensing device and that eliminates manufacturing issues associated with conductive filars. Multi-filar bundles (e.g., bifilar or trifilar) are groupings or multiple (e.g., two or three) conductors, with each conductor being surrounded by insulating coating.

Manufacturing or assembly processes that require tensioning and over-coating of a multi-filar bundle may benefit from the flex circuit assembly of the present disclosure.

[0020] Replacing embedded filars with flex circuits enables the use of conductive materials that may be formed separately and then applied to one or components of a flexible elongate member. Flex circuits allow for a larger outside diameter (OD) distal core wire and/or proximal core wire and eliminate damage/scrap due to multi-filar bundle handling. This may enable lower electrical resistance/impedance along the length of the electro-mechanical intraluminal sensing device.

Because ribbons and/or multi-filar conductor bundles add stiffness and local torques to a guidewire or catheter, the straightness and torque response of the intraluminal device with flex circuits may be improved as well, which may lead to a reduction in mechanical “whipping” responses when the device is manipulated within intravascular anatomy. Whipping occurs when the distal portion of the guidewire does not smoothly rotate with the proximal portion of the guidewire (e.g., which is controlled by a user), instead the distal portion of the guidewire rapidly rotates inside the blood vessel to catch up to the rotational position of the proximal portion of the guidewire.

[0021] Other manufacturing issues may also be alleviated by using the flex circuit assembly of the present disclosure. For example, in a current manufacturing process, a proximal end of a distal core and a distal end of a proximal core are reduced in diameter so that the multi-filar bundle may fit inside of the hypotube and allow sufficient space for the multi-filar bundle. This reduction in diameter makes it more likely for this area of the wire to kink during bending/handling. By applying the flex circuits to the outside of the distal core wire and/or the proximal core wire, the flex circuit assembly reduces or eliminates these difficulties.

[0022] Example devices incorporating a multi-filar conductor bundle and/or conductive ribbons include intraluminal medical guidewire devices as described for example in U.S. Pat. No.

10,595,820 B2, U.S. Patent Publication Nos. 2014/0187874, 2016/0058977, and 2015/0273187, and in U.S. Provisional Patent Application No. 62/552,993 (filed August 31, 2017), each of which is hereby incorporated by reference in its entirety as though fully set forth herein.

[0023] These descriptions are provided for exemplary purposes only and should not be considered to limit the scope of the metal ink conductor assembly. Certain features may be added, removed, or

modified without departing from the spirit of the claimed subject matter.

[0024] For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, and specific language will be used to describe the same. It is nevertheless understood that no limitation to the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, and methods, and any further application of the principles of the present disclosure are fully contemplated and included within the present disclosure as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one embodiment may be combined with the features, components, and/or steps described with respect to other embodiments of the present disclosure. Further, while the embodiments of the present disclosure may be described with respect to a blood vessel, it will be understood that the devices, systems, and methods described herein may be configured for use in any suitable anatomical structure or body lumen including a blood vessel, blood vessel lumen, an esophagus, eustachian tube, urethra, fallopian tube, intestine, colon, and/or any other suitable anatomical structure or body lumen. In other embodiments, the devices, systems, and methods described herein may be used to examine any number of anatomical locations and tissue types, including without limitation, organs including the liver, heart, kidneys, gall bladder, pancreas, lungs; ducts; intestines; nervous system structures including the brain, dural sac, spinal cord and peripheral nerves; the urinary tract; as well as valves within the blood vessels, chambers or other parts of the heart, and/or other systems of the body. In addition to natural structures, the device may be used to examine man-made structures such as, but without limitation, heart valves, stents, shunts, filters, and other devices. For the sake of brevity, however, the numerous iterations of these combinations will not be described separately.

[0025] FIG. 1 is a diagrammatic top view of an intraluminal device **102**, according to aspects of the present disclosure. The intraluminal device **102** may be an intravascular, intraluminal, or endoluminal device, such as a guidewire, a catheter, or a guide catheter sized and shaped for positioning within a blood vessel of a patient. The intraluminal device **102** may include a sensor **112**. For example, the sensor **112** may be a pressure sensor configured to measure a pressure of blood flow within the vessel of the patient. The intraluminal device **102** includes the flexible elongate member **106**. The sensor **112** is disposed at the distal portion **107**, also referred to as a distal subassembly, of the flexible elongate member **106**. The sensor **112** may be mounted at the distal portion **107** within a housing **280** in some embodiments. A flexible tip coil **290** extends between the housing **280** and the distal tip **108**. The connection portion **114** is disposed at the proximal portion **109**, also referred to as a proximal subassembly, of the flexible elongate member **106**. The connection portion includes the conductive portions **132**, **134**, **136**. In some embodiments, the conductive portions **132**, **134**, **136** may be conductive ink that is printed and/or deposited around the flexible elongate member **106**. In some embodiments, the conductive portions **132**, **134**, **136** may be conductive, metallic rings that are positioned around the flexible elongate member. The locking section **118** and knob or retention section **120** are disposed at the proximal portion **109** of the flexible elongate member **106**.

[0026] The intraluminal device **102** in FIG. 1 includes a distal core **210** and a proximal core **220**. The distal core **210** and the proximal core **220** are metallic components forming part of the body of the intraluminal device **102**. For example, the distal core **210** and the proximal core **220** are flexible metallic rods that provide structure for the flexible elongate member **106**. The diameter of the distal core **210** and the proximal core **220** that electrically and mechanically couples the distal core **210** to the proximal core **220** may vary along its length. A joint between the distal core **210** and proximal core **220**, which electrically and mechanically couples the distal core **210** to the proximal core **220**, is surrounded and contained by a hypotube **215**, which is a tubular member.

[0027] In some embodiments, the intraluminal device **102** includes a distal assembly and a proximal assembly that are electrically and mechanically joined together, which results in electrical

communication between the sensor **112** and the conductive portions **132, 134, 136**. For example, pressure data obtained by the sensor **112** (in this example, sensor **112** is a pressure sensor) may be transmitted to the conductive portions **132, 134, 136**. Control signals from a computer in communication with the intraluminal device **102** may be transmitted to the sensor **112** via the conductive portions **132, 134, 136**. The distal subassembly may include the distal core **210**. The distal subassembly may also include the sensor **112**, distal conductive members **230**, and/or one or more layers of polymer/plastic **240** surrounding the distal conductive members **230** and the distal core **210**. For example, the polymer/plastic layer(s) may protect the distal conductive members **230**. The proximal subassembly may include the proximal core **220**. The proximal subassembly may also include one or more layers of polymer layer(s) **250** (hereinafter polymer layer **250**) surrounding the proximal core **220** and/or conductive ribbons **260** embedded within the one or more layers of polymer layer(s) **250**. In some embodiments, the proximal subassembly and the distal subassembly may be separately manufactured. During the assembly process for the intraluminal device **102**, the proximal subassembly and the distal subassembly may be electrically and mechanically joined together. As used herein, flexible elongate member may refer to one or more components along the entire length of the intraluminal device **102**, one or more components of the proximal subassembly (e.g., including the proximal core **220**, etc.), and/or one or more components the distal subassembly (e.g., including the distal core **210**, etc.).

[0028] In various embodiments, the intraluminal device **102** may include one, two, three, or more core wires, also referred to as core members, extending along its length. For example, a single core wire may extend substantially along the entire length of the flexible elongate member **106**. In such embodiments, the locking section **118** and the knob or retention section **120** may be integrally formed at the proximal portion of the single core wire. The sensor **112** may be secured at the distal portion of the single core wire. In other embodiments, such as the embodiment illustrated in FIG. **1**, the locking section **118** and the knob or retention section **120** may be integrally formed at the proximal portion of the proximal core **220**. The sensor **112** may be secured at the distal portion of the distal core **210**. The intraluminal device **102** includes one or more distal conductive members **230** in communication with the sensor **112**. For example, the distal conductive members **230** may be one or more electrical wires that are directly in communication with the sensor **112**. In some instances, the distal conductive members **230** are electrically and mechanically coupled to and in electrical communication with the sensor **112** by, e.g., soldering. In some instances, the distal conductive members **230** include two or three electrical wires (e.g., a bifilar cable or a trifilar cable). An individual electrical wire may include a bare metallic conductor surrounded by one or more insulating layers. The distal conductive members **230** may extend along the length of the distal core **210**. For example, at least a portion of the distal conductive members **230** may be spirally wrapped in a spiral pattern around the distal core **210**.

[0029] The intraluminal device **102** includes one or more conductive ribbons **260** at the proximal portion of the flexible elongate member **106**. The conductive ribbons **260** are embedded within polymer layer(s) **250**. The conductive ribbons **260** are directly in communication with the conductive portions **132, 134**, and/or **136**. In some instances, the distal conductive members **230** are electrically and mechanically coupled to and in electrical communication with the sensor **112** by, e.g., soldering. In some instances, the conductive portions **132, 134**, and/or **136** include conductive ink (e.g., metallic nano-ink, such as silver or gold nano-ink) that is deposited or printed directed over the conductive ribbons **260**.

[0030] As described herein, electrical communication between the distal conductive members **230** and the conductive ribbons **260** may be established at the connection region **270** of the flexible elongate member **106**. By establishing electrical communication between the distal conductive members **230** and the conductive ribbons **260**, the conductive portions **132, 134, 136** may be in electrically communication with the sensor **112**.

[0031] In some embodiments represented by FIG. **1**, intraluminal device **102** includes the locking

section **118** and the knob or retention section **120**. To form the locking section **118**, a machining process is necessary to remove the polymer layer **250** and the conductive ribbons **260** in the locking section **118** and to shape proximal core **220** in the locking section **118** to the desired shape. As shown in FIG. **1**, the locking section **118** includes a reduced diameter while the knob or retention section **120** has a diameter substantially similar to that of proximal core **220** in the connection portion **114**. In some instances, because the machining process removes conductive ribbons in locking section **118**, proximal ends of the conductive ribbons **260** would be exposed to moisture and/or liquids, such as blood, saline solutions, disinfectants, and/or enzyme cleaner solutions, an insulation layer **158** is formed over the proximal end portion of the connection portion **114** to insulate the exposed conductive ribbons.

[0032] FIG. **2** is a diagrammatic side view of an intraluminal (e.g., intravascular) sensing system **100** that includes an intraluminal device **102** includes distal conductive members **230** (e.g., a multi-filar electrical conductor bundle) and conductive ribbons **260**, according to aspects of the present disclosure. The intraluminal device **102** may be an intravascular guidewire sized and shaped for positioning within a blood vessel of a patient. The intraluminal device **102** includes a distal tip **108** and a sensor **113**. For example, the sensor **113** may be a pressure sensor and/or flow sensor configured to measure a pressure of blood flow within the vessel of the patient, or another type of sensor including but not limited to a temperature or imaging sensor, or combination sensor measuring more than one property. For example, the flow data obtained by a flow sensor may be used to calculate physiological variables such as coronary flow reserve (CFR). The intraluminal device **102** includes a flexible elongate member **106**. The sensor **113** is disposed at a distal portion **107** of the flexible elongate member **106**. The sensor **113** may be mounted at the distal portion **107** within a housing **282** in some embodiments. A flexible tip coil **290** extends distally from the housing **282** at the distal portion **107** of the flexible elongate member **106**. A connection portion **114** located at a proximal end of the flexible elongate member **106** includes conductive portions **132**, **134**. In some embodiments, the conductive portions **132**, **134** may be conductive ink that is printed and/or deposited around the connection portion **114** of the flexible elongate member **106**. In some embodiments, the conductive portions **132**, **134** are conductive, may be metallic bands or rings that are positioned around the flexible elongate member. A locking area is formed by a collar or locking section **118** and knob or retention section **120** are disposed at the proximal portion **109** of the flexible elongate member **106**.

[0033] The intraluminal device **102** in FIG. **2** includes core wire including a distal core **210** and a proximal core **220**. The distal core **210** and the proximal core **220** are metallic components forming part of the body of the intraluminal device **102**. For example, the distal core **210** and the proximal core **220** may be flexible metallic rods that provide structure for the flexible elongate member **106**. The distal core **210** and/or the proximal core **220** may be made of a metal or metal alloy. For example, the distal core **210** and/or the proximal core **220** may be made of stainless steel, Nitinol, nickel-cobalt-chromium-molybdenum alloy (e.g., MP35N), and/or other suitable materials. In some embodiments, the distal core **210** and the proximal core **220** are made of the same material. In other embodiments, the distal core **210** and the proximal core **220** are made of different materials. The diameter of the distal core **210** and the proximal core **220** may vary along their respective lengths. A joint between the distal core **210** and proximal core **220** is surrounded and contained by a hypotube **215**. The sensor **113** may in some cases be positioned at a distal end of the distal core **210**.

[0034] In some embodiments, the intraluminal device **102** includes a distal subassembly and a proximal subassembly that are electrically and mechanically joined together, which creates an electrical communication between the sensor **113** and the conductive portions **132**, **134**. For example, flow data obtained by the sensor **113** (in this example, sensor **113** is a flow sensor) may be transmitted to the conductive portions **132**, **134**. In an exemplary embodiment, the sensor **113** is a single ultrasound transducer element. The transducer element emits ultrasound signals and

receives echoes. The transducer element generates electrical signals representative of the echoes. The signal carrying filars carry this electrical signal from the sensor at the distal portion to the connector at the proximal portion. The processing system **306** processes the electrical signals to extract the flow velocity of the fluid.

[0035] Control signals from a processing system **306** (e.g., a processor circuit of the processing system **306**) in communication with the intraluminal device **102** may be transmitted to the sensor **113** via a connector **314** that is attached to the conductive portions **132**, **134**. The distal subassembly may include the distal core **210**. The distal subassembly may also include the sensor **113**, the distal conductive members **230**, and/or one or more layers of insulative polymer/plastic **240** surrounding the distal conductive members **230** and the distal core **210**. For example, the polymer/plastic layer(s) may insulate and protect the conductive members of the multi-filar cable or conductor bundle. The proximal subassembly may include the proximal core **220**. The proximal subassembly may also include one or more polymer layers **250** (hereinafter polymer layer **250**) surrounding the proximal core **220** and/or conductive ribbons **260** embedded within the one or more insulative and/or protective polymer layer **250**. In some embodiments, the proximal subassembly and the distal subassembly are separately manufactured. During the assembly process for the intraluminal device **102**, the proximal subassembly and the distal subassembly may be electrically and mechanically joined together. As used herein, flexible elongate member may refer to one or more components along the entire length of the intraluminal device **102**, one or more components of the proximal subassembly (e.g., including the proximal core **220**, etc.), and/or one or more components the distal subassembly **410** (e.g., including the distal core **210**, etc.). Accordingly, flexible elongate member may refer to the combined proximal and distal subassemblies described above. The joint between the proximal core **220** and distal core **210** is surrounded by the hypotube **215**, which is a tubular member.

[0036] In various embodiments, the intraluminal device **102** may include one, two, three, or more core wires extending along its length. For example, a single core wire may extend substantially along the entire length of the flexible elongate member **106**. In such embodiments, a locking section **118** and a retention section **120** may be integrally formed at the proximal portion of the single core wire. The sensor **113** may be secured at the distal portion of the single core wire. In other embodiments, such as the embodiment illustrated in FIG. 2, the locking section **118** and the retention section **120** may be integrally formed at the proximal portion of the proximal core **220**. The sensor **113** may be secured at the distal portion of the distal core **210**. The intraluminal device **102** includes one or more distal conductive members **230** (e.g., a multi-filar conductor bundle or cable) in communication with the sensor **113**. For example, the distal conductive members **230** may be one or more electrical wires that are directly in communication with the sensor **113**. In some instances, the distal conductive members **230** are electrically and mechanically coupled to and in electrical communication with the sensor **113** by, e.g., soldering. In some instances, the distal conductive members **230** includes two or three electrical wires (e.g., a bifilar cable or a trifilar cable). An individual electrical wire may include a bare metallic conductor surrounded by one or more insulating layers. The distal conductive members **230** may extend along the length of the distal core **210**. For example, at least a portion of the distal conductive members **230** may be spirally wrapped around the distal core **210**, minimizing or eliminating whipping of the distal core within tortuous anatomy.

[0037] The intraluminal device **102** includes one or more conductive ribbons **260** at the proximal portion of the flexible elongate member **106**. The conductive ribbons **260** are embedded within polymer layer **250**. The conductive ribbons **260** are directly in communication with the conductive portions **132** and/or **134**. In some instances, the distal conductive members **230** are electrically and mechanically coupled to and in electrical communication with the sensor **113** by, e.g., soldering. In some instances, the conductive portions **132** and/or **134** includes conductive ink (e.g., metallic nano-ink, such as copper, silver, gold, or aluminum nano-ink) that is deposited or printed directed

over the conductive ribbons **260**.

[0038] As described herein, electrical communication between the distal conductive members **230** and the conductive ribbons **260** may be established at the connection portion **114** of the flexible elongate member **106**. By establishing electrical communication between the distal conductive members **230** and the conductive ribbons **260**, the conductive portions **132**, **134** may be in electrical communication with the sensor **113**.

[0039] In some embodiments represented by FIG. **1**, the intraluminal device **102** includes a locking section **118** and knob or retention section **120**. To form locking section **118**, a machining process is used to remove polymer layer **250** and conductive ribbons **260** in locking section **118** and to shape proximal core **220** in locking section **118** to the desired shape. As shown in FIG. **1**, locking section **118** includes a reduced diameter while knob or retention has a diameter substantially similar to that of proximal core **220** in the connection portion **114**. In some instances, because the machining process removes conductive ribbons in locking section **118**, proximal ends of the conductive ribbons **260** would be exposed to moisture and/or liquids, such as blood, saline solutions, disinfectants, and/or enzyme cleaner solutions, an insulation layer **158** is formed over the proximal end portion of the connection portion **114** to insulate the exposed conductive ribbons **260**.

[0040] In some embodiments, a connector **314** provides electrical connectivity between the conductive portions **132**, **134** and a patient interface monitor **304**. The Patient Interface Monitor (PIM) **304** may in some cases connect to a console or processing system **306**, which includes or is in communication with a display **308**.

[0041] The intraluminal sensing system **100** may be deployed in a catheterization laboratory having a control room. The processing system **306** may be located in the control room. Optionally, the processing system **306** may be located elsewhere, such as in the catheterization laboratory itself. The catheterization laboratory may include a sterile field while its associated control room may or may not be sterile depending on the procedure to be performed and/or on the health care facility. In some embodiments, the intraluminal device **102** may be controlled from a remote location such as the control room, such that an operator is not required to be in close proximity to the patient.

[0042] The intraluminal device **102**, PIM **304**, and display **308** may be communicatively coupled directly or indirectly to the processing system **306**. These elements may be communicatively coupled to the processing system **306** via a wired connection such as a standard copper multi-filar conductor bundle. The processing system **306** may be communicatively coupled to one or more data networks, e.g., a TCP/IP-based local area network (LAN). In other embodiments, different protocols may be utilized such as Synchronous Optical Networking (SONET). In some cases, the processing system **306** may be communicatively coupled to a wide area network (WAN).

[0043] The PIM **304** transfers the received signals to the processing system **306** where the information is processed and displayed (e.g., as physiology data in graphical, symbolic, or alphanumeric form) on the display **308**. The console or processing system **306** may include a processor and a memory. The processing system **306** may be operable to facilitate the features of the intraluminal sensing system **100** described herein. For example, the processor may execute computer readable instructions stored on the non-transitory tangible computer readable medium.

[0044] The PIM **304** facilitates communication of signals between the processing system **306** and the intraluminal device **102**. The PIM **304** may be communicatively positioned between the processing system **306** and the intraluminal device **102**. In some embodiments, the PIM **304** performs preliminary processing of data prior to relaying the data to the processing system **306**. In examples of such embodiments, the PIM **304** performs amplification, filtering, and/or aggregating of the data. In an embodiment, the PIM **304** also supplies high-and low-voltage DC power to support operation of the intraluminal device **102** via the distal conductive members **230**.

[0045] A multi-filar cable or transmission line bundle, such as distal conductive members **230**, may include a plurality of conductors, including one, two, three, four, five, six, seven, or more conductors. In the example shown in FIG. **2**, the multi-filar conductor bundle includes two straight

portions **232** and **236**, where the multi-filar conductor bundle lies parallel to a longitudinal axis of the flexible elongate member **106**, and a spiral portion **234**, where the multi-filar conductor bundle is wrapped around the exterior of the flexible elongate member **106** and then overcoated with an insulative and/or protective polymer **240**. Communication, if any, along the multi-filar conductor bundle may be through numerous methods or protocols, including serial, parallel, and otherwise, where one or more filars of the of the multi-filar conductor bundle carry signals. One or more filars of the multi-filar conductor bundle may also carry direct current (DC) power, alternating current (AC) power, or serve as a ground connection.

[0046] The display or monitor **308** may be a display device such as a computer monitor or other type of screen. The display or monitor **308** may be used to display selectable prompts, instructions, and visualizations of imaging data to a user. In some embodiments, the display **308** may be used to provide a procedure-specific workflow to a user to complete an intraluminal imaging procedure.

[0047] Before continuing, it should be noted that the examples described above are provided for purposes of illustration and are not intended to be limiting. Other devices and/or device configurations may be utilized to carry out the operations described herein.

[0048] FIG. **3A** is a schematic top view of an unrolled flex circuit **320** for application to a portion of the intraluminal device **102** with electrical conductors/traces **328** formed in the flex circuit **320**, according to aspects of the present disclosure. In some embodiments, the flex circuit **320** may be formed in a separate process before being applied to a distal core **210**. The flex circuit **320** is a thin film that includes a flexible substrate **322** onto or within which a plurality of electrical conductors/traces **328** are formed. In some embodiments, the flex circuit **320** may be formed by disposing the electrical conductors/traces **328** onto a polymer base and then overcoating the electrical conductors/traces **328** with a polymer topcoat.

[0049] The flexible substrate **322** may be constructed to include a film layer of a flexible polyimide material such as KAPTON™ (trademark of DuPont). Other suitable materials include polyester films, polyimide films, polyethylene naphthalate films, or polyetherimide films, liquid crystal polymer, other flexible printed semiconductor substrates as well as products such as Upilex® (registered trademark of Ube Industries) and TEFLON® (registered trademark of E.I. du Pont). In the flat configuration illustrated in FIG. **3A**, the flexible substrate **322** has a generally rectangular shape. As shown and described herein, the flexible substrate **322** is configured to be wrapped partially or completely around an exterior the distal core **210** in some instances. Therefore, the thickness of the film layer of the flexible substrate **322** is generally related to the degree of curvature in the final assembled flexible elongate member **106**. In some embodiments, the film layer is between 5 μm and 100 μm, with some particular embodiments being between 5 μm and 25.1 μm, e.g., 6 μm.

[0050] To electrically interconnect the conductive members **231** and the conductive member **261**, in an embodiment, the flexible substrate **322** includes conductive traces **328** formed in the film layer that carry signals between the conductive members **231** and the conductive members **261**. In particular, the conductive traces **328** provide communication between the conductive members **231** and the conductive members **261** extending along the flexible substrate **322**. In some instances, the conductive traces **328** may also facilitate electrical communication between the PIM **304** and the sensor **112/113**. The conductive traces **328** may also provide a set of conductive pads **324** and **326** that contact the conductive members **231** and the conductive members **261**, respectively, when the conductive members **231** and the conductive members **261** are mechanically and electrically coupled to the flexible circuit **320**. Suitable materials for the conductive traces **328** include copper, gold, aluminum, silver, tantalum, nickel, and tin, and may be deposited on the flexible substrate **322** by processes such as sputtering, plating, and etching. In an embodiment, the flexible substrate **322** includes a chromium adhesion layer. The width and thickness of the conductive traces **328** are selected to provide proper conductivity and resilience when the flexible substrate **322** is rolled. In that regard, an exemplary range for the thickness of a conductive traces **328** and/or conductive pad

is between 1-5 μm . For example, in an embodiment, 5 μm conductive traces **328** are separated by 5 μm of space. The width of a conductive trace **328** on the flexible substrate may be further determined by the width of the conductive members **231** and the conductive members **261** to be coupled to the trace/pad.

[0051] In some embodiments, the electrical conductors/traces **328** are distinct electrical traces in that they are physically and electrically separate from one another, but structurally made of a same material. In other embodiments, the electrical conductors/traces **328** may structurally be made of different materials. On a distal end of the flexible substrate **322**, a plurality of distal conductive pads **324** are formed and are electrically and mechanically coupled to respective ones of the plurality of electrical conductors/traces **328** by, e.g., soldering, ultrasonic welding, conductive adhesive, etc. Additionally, on a proximal end of the flexible substrate **322**, a plurality of proximal conductive pads **326** are formed and are electrically and mechanically coupled to respective ones of the plurality of electrical conductors/traces **328** by, e.g., soldering, ultrasonic welding, conductive adhesive, etc.

[0052] In some embodiments, the flex circuit **320** may be in the form of a sheet that is wrapped around an exterior of the distal core **210** along a longitudinal axis such that edges of the flex circuit **320** meet. That is, the flex circuit **320** extends parallel to a longitudinal axis of the flexible elongate member **106** over the distal core **210**. In some embodiments, the flex circuit **320** may be in the form of a strip such that the strip is spirally wrapped around the distal core **210**. Once applied to the distal core **210**, electrical connections are made to conductive members **231** and conductive members **261**. In some embodiments, conductive members **231** and conductive members **261** may be multi-filar cables. In some embodiments, conductive members **261** may be conductive ribbons, such as conductive ribbons **260**. In some embodiments, respective ones of conductive members **231** are electrically and mechanically coupled to respective ones of the plurality of distal conductive pads **324** by, e.g., soldering, ultrasonic welding, conductive adhesive, etc. In some embodiments, respective ones of conductive members **261** are electrically and mechanically coupled to respective ones of the plurality of proximal conductive pads **326** by, e.g., soldering, ultrasonic welding, conductive adhesive, etc. In some embodiments, respective ones of conductive ribbons **260** are directly coupled to the respective ones of the plurality of proximal conductive pads **326**. That is, conductive ribbons **260** may be formed past the connection region **270** and the polymer layer **250** in which the conductive ribbons **260** are embedded is removed such the conductive ribbons **260** may be directly coupled to respective ones of the plurality of proximal conductive pads **326**. In some embodiments, respective ones of conductive ribbons **260** may be indirectly coupled to the respective ones of the plurality of proximal conductive pads **326**. That is, respective distal ends of the conductive members **261** may be coupled to respective ones of the plurality of proximal conductive pads **326** and respective proximal ends of the conductive members **261** may be coupled to respective ones of the conductive ribbons **260** the connection region **270**.

[0053] Aspects of the present disclosure may include features described in App. No. _____, filed _____, and titled “Electrical Traces Along Core Wire For Intraluminal Physiology Sensing Guidewire And Associated Devices, Systems, And Methods” (Atty Dkt No. 2021PF00918/44755.2265PV01), App. No. _____, filed _____, and titled “Flex Circuit For Electrical Connection In Intraluminal Device And Associated Devices, Systems, And Methods” (Atty Dkt No. 2021PF00919/44755.2266PV01), and App. No. _____, filed _____, and titled “Continuous Electrical Trace In Intraluminal Device And Associated Devices, Systems, And Methods” (Atty Dkt No. 2021PF00897/44755.2270PV01), the entirety of which are hereby incorporated by reference herein.

[0054] The intraluminal device **102** may include any suitable quantity of conductors in the flex circuit **320**, e.g., two, three, four, five, or more. In some embodiments, the intraluminal device **102** includes the same quantity of conductors in the flex circuit **320** as the distal conductive members **230**. In some embodiments, a communication line is established between one of conductive

members **231**, electrical conductors/traces **328**, and one of conductive members **261** that are electrically coupled/in communication with one another. In some embodiments, there are two or more communication lines for, e.g., power (positive, negative, ground), signal/data transmission. [0055] FIG. **3B** is a schematic end view of a rolled flex circuit **320** for application to a portion of the intraluminal device **102** from direction P shown in FIG. **3A** rolled in a cylindrical configuration along longitudinal axis **340** and a cross-sectional end view along section line A-A in FIG. **5**, according to aspects of the present disclosure. FIG. **3B** is not a complete cross-sectional end view of FIG. **5**, because FIG. **3** does not include all of components in the section of FIG. **5**. That is, FIG. **3B** shows a subset of the components including distal core **210**, adhesive **332**, flex circuit **320**, and proximal conductive pads **326**, FIG. **5** shows that subset and additionally shows polymer coating **364** and hydrophilic coating **366**. In the depicted embodiment illustrated in FIG. **3B**, the flex circuit **320** is wrapped around, in this instance, the distal core **210**. Flex circuit **320** surrounds and couples to the distal core **210** via adhesive **332**. That is, adhesive **332** is in contact with the distal core **210** and flex circuit **320** is in contact with adhesive **332**. Also visible from direction P, are the plurality of proximal conductive pads **326**.

[0056] FIG. **4A** is a schematic side view of a portion of the flex circuit **320** including the electrical conductors/traces **328** disposed onto the flexible substrate **322** of the flex circuit **320**, according to aspects of the present disclosure. That is, the flex circuit **320** is a thin film that includes the flexible substrate **322** onto which, in this illustration, the electrical conductors/traces **328** are disposed. On a distal end of the flexible substrate **322**, a distal conductive pad **324** is formed. On a proximal end of the flexible substrate **322**, a proximal conductive pad **326** is formed on a same side of the flexible substrate **322** that the distal conductive pad **324** is formed.

[0057] FIG. **4B** is a schematic side view of a portion of the flex circuit **320** including the electrical conductors/traces **328** disposed within the flexible substrate **322** of the flex circuit **320**, according to aspects of the present disclosure. In this illustration, the electrical conductors/traces **328** are formed within the substrate by, for example, forming a portion of the substrate (e.g., a polymer base **323**), disposing the electrical conductors/traces **328** (e.g., on the polymer base **323**), and then forming the rest of the flexible substrate **322** over the electrical conductors/traces **328** (e.g., the polymer topcoat **325**). Proximal conductive pad **326** may be formed over the polymer coating **364**. In this illustration, the distal conductive pad **324** and the proximal conductive pad **326** are formed on a same side of the flexible substrate **322**.

[0058] FIG. **4C** is a schematic side view of a portion of the flex circuit **320** including the electrical conductors/traces **328** disposed within the flexible substrate **322** of the flex circuit **320**, according to aspects of the present disclosure. In this illustration, the distal conductive pad **324** and the proximal conductive pad **326** are formed on opposite sides of the flexible substrate **322**.

[0059] FIG. **4D** is a schematic side view of a portion of the flex circuit **320** including the electrical conductors/traces **328** disposed within the flexible substrate **322** of the flex circuit **320**, according to aspects of the present disclosure. In this illustration, the distal conductive pad **324** and the proximal conductive pad **326** are formed in the ends of the flexible substrate **322**.

[0060] FIG. **5** is a diagrammatic side view of a portion of an intraluminal device **102** with electrical conductors/traces **328** formed in the flex circuit **320**, according to aspects of the present disclosure. Visible is the flex circuit **320** is joined to the distal core **210** by adhesive **332**. That is, adhesive **332** is in contact the distal core **210** and flex circuit **320** is in contact with the adhesive **332**. While FIG. **5** illustrates adhesive **332** along the entire length of the wider portion of distal core **210**, in some embodiments, adhesive **332** may be used only at the ends of the flex circuit **320**. In some embodiments, adhesive **332** may be spaced apart, such as every 3 cm, 5 cm, etc. The flex circuit **320** is a thin film that includes a flexible substrate **322** onto or within which a plurality of electrical conductors/traces **328** are formed. In embodiments where adhesive **332** is not used along the entire length of the wider portion of distal core **210**, the flexible substrate **322** of the flex circuit **320** may directly contact the distal core **210**.

[0061] The flex circuit **320** may be positioned around a majority of the length of the distal core. In some embodiments, the distal core **210** may be between 30 cm and 50 cm, e.g. 40 cm. In some embodiments, the proximal core **220** may be between 130 cm and 160 cm, e.g. 145 cm. In some embodiments, the flex circuit **320** may be between 20 cm and 60 cm, between 20 cm and 40 cm, etc., e.g. 30-40 cm. Therefore, for example, the flex circuit **320** may be between 50% and 100% of the length of the distal core **210**, between 66% and 100% of the length of the distal core **210**, between 75% and 100% of the length of the distal core **210**, etc. In some embodiments, the length of the flex circuit **320** is less than the length of the distal core **210**. In some embodiments, the length of the flex circuit **320** is equal to length of distal core **210**. In some embodiments, the length of the flex circuit **320** may be longer than the length of the distal core **210**. The flex circuit **320** may be longitudinally centered relative to the distal core **210**. The flex circuit **320** may be positioned closer to the distal end of the distal core **210** (e.g., distal ends of distal core **210** and flex circuit **320** are aligned or spaced apart from one another). The flex circuit **320** may be positioned closer to the proximal end of the distal core **210** (e.g. . . . , proximal ends of distal core **210** and flex circuit **320** are aligned or spaced apart from one another). In some embodiments, all of the length of the flex circuit **320** is positioned around the distal core **210**. In some embodiments, a portion of the flex circuit **320** may not be positioned around the distal core **210**. For example, a distal end of the flex circuit **320** may be distal of the distal end of the distal core **210**. For example, a proximal end of the flex circuit **320** may be proximal of the proximal end of the distal core **210**.

[0062] In some instances, on a distal end of the flex circuit **320**, a plurality of distal conductive pads **324** are formed and are electrically and mechanically coupled to respective ones of the plurality of electrical conductors/traces **328** by, e.g., soldering, ultrasonic welding, conductive adhesive, etc. In that regard, respective proximal ends of conductive members **231** are coupled to respective ones of the plurality of distal conductive pads **324** by a coupling **362**, e.g., soldering, ultrasonic welding, conductive adhesive, etc. Respective distal ends of conductive members **231** are coupled to respective pads of sensor **112** by, e.g., soldering, ultrasonic welding, conductive adhesive, etc.

[0063] In some instances, the flex circuit **320** directly couples (e.g., mechanically and/or electrically connects) to the sensor **112**. For example, a distal portion of the flex circuit **320** can overlap along a length of the intraluminal device with a portion of the sensor **112** including conductive pads. In some instances, the distal portion of the flex circuit **320** (with conductive pads **324**) is positioned over the proximal portion of the sensor **112** (with conductive pads of the sensor **112**). The conductive pads of the sensor **112** receive electrical signals generated by the sensor **112** (e.g., a pressure-sensing diaphragm, a flow-sensing ultrasound transducer, etc.) Conductive pads **324** of the flex circuit **320** (at the distal portion of the flex circuit **320**) can directly couple (e.g., mechanically and/or electrically connect) to conductive pads of the sensor **112**. Accordingly, the sensor **112** is in electrical communication with the plurality of electrical conductors/traces **328** of the flex circuit **320** via the coupling between the conductive pads of the flex circuit **320** and the conductive pads **324** of the flex circuit. In such instances, conductive members **231** are omitted so that there is no intermediate conductive member between the flex circuit **320** and the sensor **112**.

[0064] On a proximal end of the flex circuit **320**, a plurality of proximal conductive pads **326** are formed and are electrically and mechanically coupled to respective ones of the plurality of electrical conductors/traces **328**. Respective ones of conductive members **261** are coupled to the respective ones of the plurality of proximal conductive pads **326** by a coupling **362**, e.g., soldering, ultrasonic welding, conductive adhesive, etc. In some embodiments, conductive members **261** may couple to conductive ribbons **260** (also referred to as proximal conductive members **260**) associated with the proximal core **220** via a multi-filar cable that runs on an exterior or an interior of hypotube **215**. A polymer coating **364** is then applied, which is then overcoated with hydrophilic coating **366**.

[0065] FIG. **6** is a diagrammatic cross-sectional side view of a portion of an intraluminal device **102** with electrical conductors/traces **328** formed in the flex circuit **320**, according to aspects of the

present disclosure. FIG. 6 includes features similar to those described in FIG. 5. In the embodiment of FIG. 6, in section 600, the flex circuit 320 replaces the polymer coating 364. In some embodiments, replacing the polymer coating 364 in section 600 with the flex circuit 320 allows for an increased diameter of distal core 210. In order for flex circuit 320 to replace the polymer coating 364, the distal conductive pad 324 and the proximal conductive pad 326 are formed in the ends of the flexible substrate 322. As with FIG. 5, in FIG. 6 the flex circuit 320 is joined to the distal core 210 by adhesive 332. That is, adhesive 332 is in contact the distal core 210 and flex circuit 320 is in contact with the adhesive 332. While FIG. 6 illustrates adhesive 332 along the entire length of the wider portion of distal core 210, in some embodiments, adhesive 332 may be used only at the ends of the flex circuit 320. In some embodiments, adhesive 332 may be spaced apart, such as every 3 cm, 5 cm, etc. In embodiments where adhesive 332 is not used along the entire length of the wider portion of distal core 210, the flexible substrate 322 of the flex circuit 320 may directly contact the distal core 210. The flex circuit 320 is then overcoated with hydrophilic coating 366.

[0066] In some instances, the flex circuit 320 directly couples (e.g., mechanically and/or electrically connects) to the sensor 112. For example, a distal portion of the flex circuit 320 can overlap along a length of the intraluminal device with a portion of the sensor 112 including conductive pads. In some instances, the distal portion of the flex circuit 320 (with conductive pads 324) is positioned over the proximal portion of the sensor 112 (with conductive pads of the sensor 112). The conductive pads of the sensor 112 receive electrical signals generated by the sensor 112 (e.g., a pressure-sensing diaphragm, a flow-sensing ultrasound transducer, etc.) Conductive pads 324 of the flex circuit 320 (at the distal portion of the flex circuit 320) can directly couple (e.g., mechanically and/or electrically connect) to conductive pads of the sensor 112. Accordingly, the sensor 112 is in electrical communication with the plurality of electrical conductors/traces 328 of the flex circuit 320 via the coupling between the conductive pads of the flex circuit 320 and the conductive pads 324 of the flex circuit. In such instances, conductive members 231 are omitted so that there is no intermediate conductive member between the flex circuit 320 and the sensor 112.

[0067] FIG. 7 is a diagrammatic cross-sectional side view of a portion of an intraluminal device 102 with electrical conductors/traces 328 formed in the flex circuit 320, according to aspects of the present disclosure. FIG. 7 includes features similar to those described in FIG. 6. In the embodiment in FIG. 7, the flex circuit 320 is applied along a length of the distal core 210 and the proximal core 220, replacing not only the polymer coating 364 but also the hypotube 215. In some embodiments, replacing the polymer coating 364 along a length of the distal core 210 and the proximal core 220 allows for an increased diameter of not only the distal core 210 but also the proximal core 220. In order for flex circuit 320 to replace the polymer coating 364, the distal conductive pad 324 and the proximal conductive pad 326 are formed in the ends of the flexible substrate 322. In some embodiments, the flex circuit 320 is joined to the distal core 210 and the proximal core 220 by adhesive 332. That is, adhesive 332 is in contact the distal core 210 and the proximal core 220 and the flex circuit 320 is in contact with the adhesive 332. While FIG. 7 illustrates adhesive 332 along the entire length of the distal core 210 and the proximal core 220, in some embodiments, adhesive 332 may be used only at the ends of the flex circuit 320. In some embodiments, adhesive 332 may be spaced apart, such as every 3 cm, 5 cm, etc. In embodiments where adhesive 332 is not used along the entire length of the wider portion of distal core 210, the flexible substrate 322 of the flex circuit 320 may directly contact the distal core 210. The flex circuit 320 is then overcoated with hydrophilic coating 366.

[0068] In some instances, the flex circuit 320 directly couples (e.g., mechanically and/or electrically connects) to the sensor 112. For example, a distal portion of the flex circuit 320 can overlap along a length of the intraluminal device with a portion of the sensor 112 including conductive pads. In some instances, the distal portion of the flex circuit 320 (with conductive pads 324) is positioned over the proximal portion of the sensor 112 (with conductive pads of the sensor 112). The conductive pads of the sensor 112 receive electrical signals generated by the sensor 112

(e.g., a pressure-sensing diaphragm, a flow-sensing ultrasound transducer, etc.) Conductive pads **324** of the flex circuit **320** (at the distal portion of the flex circuit **320**) can directly couple (e.g., mechanically and/or electrically connect) to conductive pads of the sensor **112**. Accordingly, the sensor **112** is in electrical communication with the plurality of electrical conductors/traces **328** of the flex circuit **320** via the coupling between the conductive pads of the flex circuit **320** and the conductive pads **324** of the flex circuit. In such instances, conductive members **231** are omitted so that there is no intermediate conductive member between the flex circuit **320** and the sensor **112**. [0069] Accordingly, it may be seen that the flex circuit assembly advantageously enables the use of conductive materials that may be formed separately and then applied to one or components of a flexible elongate member. Additionally, flex circuits allow for a larger outside diameter (OD) of the core wires and eliminate damage/scrap due to multi-filar bundle handling and thus, may enable lower electrical resistance/impedance along the length of the electro-mechanical intraluminal sensing device. A straightness and torque response of the intraluminal device may also be improved because ribbons and/or multi-filar conductor bundles, which would be reduced or eliminated, add stiffness and local torques to a guidewire or catheter. A number of variations are possible on the examples and embodiments described above. The flex circuit assembly could be applied to any product that involves electrical conductors to be embedded into composite subassemblies. However, unlike conductive ribbons, flex circuits may conform to the curvature of the core wire. [0070] The logical operations making up the embodiments of the technology described herein are referred to variously as operations, steps, objects, elements, components, or modules. Furthermore, it should be understood that these may be arranged or performed in any order, unless explicitly claimed otherwise or a specific order is inherently necessitated by the claim language. It should further be understood that the described technology may be employed in single-use and multi-use electrical and electronic devices for medical or nonmedical use. [0071] All directional references e.g., upper, lower, inner, outer, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, proximal, and distal are only used for identification purposes to aid the reader's understanding of the claimed subject matter, and do not create limitations, particularly as to the position, orientation, or use of the metal ink conductor assembly. Connection references, e.g., attached, coupled, connected, and joined are to be construed broadly and may include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily imply that two elements are directly connected and in fixed relation to each other. The term “or” shall be interpreted to mean “and/or” rather than “exclusive or.” The word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. Unless otherwise noted in the claims, stated values shall be interpreted as illustrative only and shall not be taken to be limiting. [0072] The above specification, examples and data provide a complete description of the structure and use of exemplary embodiments of the metal ink conductor assembly as defined in the claims. Although various embodiments of the claimed subject matter have been described above with a certain degree of particularity, or with reference to one or more individual embodiments, those skilled in the art could make numerous alterations to the disclosed embodiments without departing from the spirit or scope of the claimed subject matter. [0073] Still other embodiments are contemplated. It is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative only of particular embodiments and not limiting. Changes in detail or structure may be made without departing from the basic elements of the subject matter as defined in the following claims.

Claims

- 1.** An intraluminal device, comprising: a flexible elongate member configured to be positioned within a body lumen of a patient, wherein the flexible elongate member comprises: a distal portion comprising a distal core member and a sensor, wherein the distal core member comprises a length between a distal end and a proximal end; a proximal portion comprising a proximal electrical conductor; and a flex circuit positioned around a majority of the length of the distal core member, wherein the flex circuit comprises a distal portion coupled to the sensor and a proximal portion coupled to a proximal electrical conductor such that the proximal electrical conductor is in electrical communication with the sensor via the flex circuit.
- 2.** The intraluminal device of claim 1, wherein the flex circuit comprises: a polymer base; and a conductive trace formed on the polymer base.
- 3.** The intraluminal device of claim 2, wherein the flex circuit further comprises a polymer topcoat formed over the conductive trace.
- 4.** The intraluminal device of claim 2, wherein the flex circuit further comprises: a distal conductive pad formed on the polymer base at a distal portion of the flex circuit; and a proximal conductive pad formed on the polymer base at a proximal portion of flex circuit, wherein the conductive trace is coupled to the distal conductive pad and the proximal conductive pad, wherein the proximal electrical conductor is coupled to the flex circuit at the proximal conductive pad, and wherein the sensor is coupled to the flex circuit at the distal conductive pad.
- 5.** The intraluminal device of claim 1, further comprising an adhesive positioned between the flex circuit and the distal core member, wherein the adhesive is configured to couple the flex circuit and the distal core member.
- 6.** The intraluminal device of claim 1, wherein the flex circuit is wrapped in a spiral pattern around the distal core member.
- 7.** The intraluminal device of claim 1, wherein the flex circuit is rolled in a cylindrical configuration around the distal core member.
- 8.** The intraluminal device of claim 1, wherein the flex circuit extends parallel to a longitudinal axis of the flexible elongate member around the distal core member.
- 9.** The intraluminal device of claim 1, wherein the flex circuit is formed prior to being positioned around the distal core member.
- 10.** The intraluminal device of claim 9, wherein the flex circuit is formed as a sheet, wherein at least one conductive trace of the flex circuit is formed across the sheet, and wherein the sheet is wrapped around the distal core member.
- 11.** The intraluminal device of claim 1, wherein the proximal portion further comprises a proximal core member, wherein the proximal core member comprises a distal end and a proximal end, wherein the flex circuit is positioned around the proximal core member.
- 12.** The intraluminal device of claim 1, wherein the distal portion further comprises: a polymer coating positioned around the flex circuit.
- 13.** The intraluminal device of claim 12, wherein the polymer coating is in direct contact with the flex circuit.
- 14.** The intraluminal device of claim 1, wherein the distal portion further comprises: a hydrophilic coating positioned around the flex circuit.
- 15.** The intraluminal device of claim 14, wherein the hydrophilic coating is in direct contact with the flex circuit.
- 16.** An apparatus, comprising: an intravascular guidewire configured to be positioned within a blood vessel of a patient, the intravascular guidewire comprising: a distal portion comprising a distal core member and a sensor, wherein the distal core member comprises a length between a distal end and a proximal end; a proximal portion comprising a proximal electrical conductor; and a flex circuit positioned around a majority of the length of the distal core member, wherein the flex circuit comprises a distal portion coupled to the sensor and a proximal portion coupled to a

proximal electrical conductor such that the proximal electrical conductor is in electrical communication with the sensor via the flex circuit.
